Electrostatically Actuated Robotic Fish: Design and Control for High-Mobility Open-Loop Swimming

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***Abstract*—This paper presents a project that aims at fabricating a biologically inspired robotic fish. The robotic fish is designed to be capable of propelling itself through oscillations of a flexible caudal fin, like a real underwater fish. In this paper, we describe the design features that underlie the operation of the robotic fish. These features include a unique actuator referred to as electrostatic film motor and a light and flexible power transmission system. The electrostatic film motor is made of two pieces of flexible printed circuit film and can be utilized as a new-type artificial muscle. The power transmission system permits reciprocating power to be converted to periodic oscillations and distributed to the caudal fin. Based on several design considerations inspired by biological concepts, we propose several open-loop swimming control strategies for the constructed robotic fish to accomplish fish-like motion (i.e., cruising, turning, and diving). Experiments of Seidengyo I, the first prototype of our electrostatic fish family, are carried out to confirm the validity of the original design and control. We further design Seidengyo II to improve on Seidengyo I and show the results of the experiments.**

***Index Terms*—Artificial muscle, biomimetic robotic fish, electrostatic film motor, fish-like swimming, open-loop control.**

I. Introduction

MANY previous studies of submersible vehicles have been performed, including studies on conventional autonomous underwater vehicles relying on small screw propellers and on fish-like robots relying on fin-based motion for the forward thrust. Since fish are skilful in physical movements and power-efficient endurance swimmers, the better route toward designing submersible vehicles with similar capabilities is to understand fish physiology and control strategies. Then, more researches have been devoted to the hydrodynamics mechanism of fish-like swimming [1], the different maneuverable swimming modes [2], and the mechanical structure of fish-like robot models [3].

We believe that the correct concept of biomimetic designs is to distill the fundamental principles of effective fish performance, and then, apply them to the design of robotic fish. For example, fish use muscles to actuate their propulsors. But, muscles have a surprising variety of functions in locomotion, serving as motors, brakes, springs, and struts [4]. It is inadequate and impossible to entirely duplicate these functions of muscles due to their complexity and redundancy. To date, a variety of artificial muscles, as new-style actuators, have been already explored and applied to robotic fish [5]–[8].

Our proposition is to realize fish-like mobility using the oscillation of a flexible caudal fin and make the robotic fish more like natural fish. This proposition is made possible by a light, thin, and flexible actuator called electrostatic film motor that is originally developed in our laboratory. As a matter of fact, electrostatic motors were developed as microelectromechanical systems (MEMS) actuators because of their planar structures and high power densities at micro scale [9], [10]. As some further structure studies have progressed in recent years [11]–[14], it has become generally accepted that the electrostatic film motor can be applied to not only micromachining but also to ordinary-sized devices such as robotic servo actuators. Inspired by these previous studies, we take advantage of the unique actuator to propel our robotic fish. To the best of our knowledge, our designed robot is the first attempt where an electrostatic film motor is used as an artificial muscle in a robotic fish.

In this paper, we first describe an original two- to four-phase electrostatic film motor and briefly analyze its dynamics. This analysis provides insight into the basic property of an open-loop control system capable of easily tuning displacement and velocity of the unique actuator. Next, we utilize the unique actuator to fabricate the Seidengyo I robotic fish (i.e., the first prototype in the electrostatic fish family) and propose several swimming control strategies. We then demonstrate the validity of the design and control through experimental results of Seidengyo I. Finally, aiming at solving weak points of Seidengyo I, we design a Seidengyo II robotic fish to improve its adaptability and maneuverability. Furthermore, the fundamental contribution of our robotic fish is the demonstration that fast and stable fish-like locomotion can be achieved with an open-loop actuation pattern. This suggests that the mechanical impedance from the stiffness and viscosity of rhythmically active actuator can allow rejection of rapid perturbations and simplify control [15]. Such insight concerning passive stabilization will facilitate the design of an autonomous sensory feedback. We consider that our robotic fish will represent a relevant contribution in bioinspired

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Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

This paper has supplementary downloadable material available at http://ieeexplore.ieee.org, provided by the authors. It comprises four videos, converted to Microsoft MPEG-1 mode that can be played in Windows Media Player. The videos are of sizes 7.32 MB, 12.7 MB, 5.7 MB, and 21.2 MB, which show a 2–4-phase electrostatic films motor, Seidengyo I, the first prototype of robotic fish family, imitation of fish's muscle function, and Seidengyo II whose body forms a streamline shape.

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robotics research and has even potential applications in education and entertainment.

II. Biological Inspiration

Natural selection has ensured that animals' musculoskeletals evolve into optimal and high-efficient mechanical systems. Then, much recent interest in the field of fish swimming robots has been placed on the adoption of principlesxg found in animal locomotion [16], [17]. To accomplish fast and robust swimming, we also draw design inspiration from actual fish. On the one hand, we study the fish's morphology and swimming mode, and abstract a remarkable approach from them. On the other hand, we introduce a new-style artificial muscle (i.e., electrostatic film motor) by understanding fundamental principle of fish muscles.

A. Fish Locomotion

Fish mainly adopt two kinds of swimming locomotion: 1) body and/or caudal fin (BCF) locomotion and 2) median and/or paired fin (MPF) locomotion [18]. Sfakiotakis *et al.* [16] further indicated that the MPF mode rarely contribute to forward propulsion. As a matter of fact, a great number of fish rely on the BCF mode for propulsion and on the MPF mode for maneuvering and stabilization [18]. Moreover, the both swimming locomotion generally derive from two types of essential movement characteristics: undulatory motion and oscillatory motion. Since oscillatory motion can eventually be generated via the gradual increase of the undulation wavelength, the two types of motion can be considered to be a continuum. This suggests that the generating mechanisms of the two motions are almost identical. Namely, both motions result from the coupled oscillation of small elements (e.g., muscles). In the field of biology, the fish swimming is classified as five modes (i.e., anguilliform, subcarangiform, carangiform, thunniform, and ostraciiform) according to the different undulation wavelength [19]. In these modes, the thunniform mode utilized by tuna, shark, and marine mammals is the most efficient locomotion mode, where significant lateral movements occur only at the caudal fin and at the area near the narrow peduncle. In addition, thunniform swimmers have low-drag streamline body shapes, narrow peduncles, and tall lunate caudal fins. The design is significantly suitable for high-speed cruise in calm water [16].

Motivated by the review from biology, many researches concerning robotic fish have focused their attention on the locomotion mode like tuna and sharks [1], [20], [21]. In the first stage of our development, our robotic fish will also use the thunniform swimming and neglect motions of median and paired fins because our basic idea for realizing the fish-like locomotion is to only utilize one actuator to generate propulsion.

B. New-Style Artificial Muscle

As mentioned in [18], fish swimming is characterized by the activation of the musculature on alternate sides of the body. In fishes that generate the bulk of thrust with their caudal fins, anterior muscles generate power, which is transmitted to the fin in part through the stiffening action of posterior muscles [4].

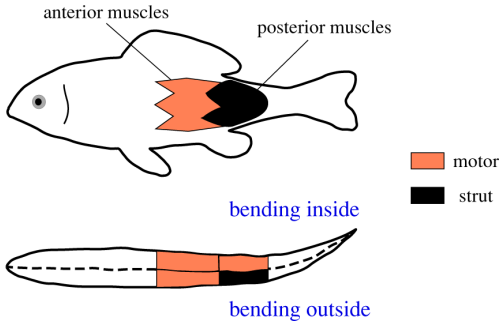


Fig. 1. In fish, axial muscle may play either a force-generating or a force-transmitting role, depending on when it is activated with respect to undulatory wave (adapted from [22]).

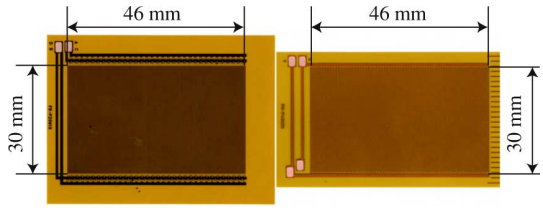


Fig. 2. Electrostatic film motor consists of two pieces of FPC boards: stator (left) and slider (right).

Here, muscles play not only a force-generating but also a force-transmitting role. As shown in Fig. 1, the function of muscles varies within a caudal fin oscillation cycle. The anterior muscles at the bending outside shorten and produce power, which is transmitted by posterior muscles acting as struts. In contrast, all muscles at the bending inside act as motors.

Since muscles have relatively complex constructions and performances, it is difficult to provide finished artificial muscles as substitutes for them. In general, electromagnetic motor is a normal selection as an artificial muscle for providing the propulsion while a robotic fish is designed [23]–[25]. However, Barrett *et al.* [26] indicated that the actuation made of electromagnetic motors is noisy and requires a fairly elaborate transmission system to amplify torque. Thus, many other actuators (e.g., shape memory alloy actuators, piezoelectric actuators, and conducting polymer actuators) have been studied and applied to robotic fish as artificial muscles [3], [5]–[8].

In this paper, we do not discuss these existent artificial muscles since the subject is beyond the scope of this paper. Conversely, we focus on a unique actuator referred to as electrostatic film motor, which is made of flexible printed circuit (FPC) films (see Fig. 2).

III. Description of Electrostatic Film Motor

As plotted in Fig. 3, the electrostatic film motor is a synchronous motor with six phases and utilizes film electrodes for both stator and slider. While the slider film is directly placed on the stator film, a planar, light, flexible, and high-power linear motor is constructed. The slider film is driven by the electrostatic force so as to accomplish the linear motion. It is important to note that the slider is excited by two-phase ac voltage and the stator is excited by four-phase ac voltage, differing from two

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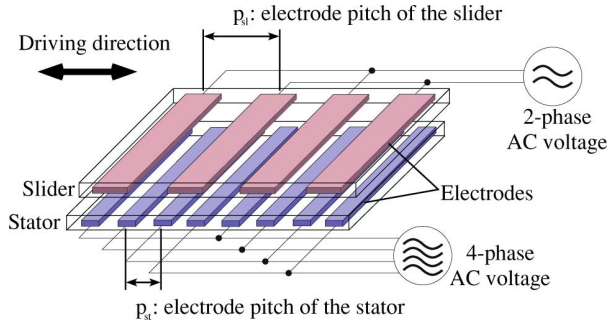


Fig. 3. Electrostatic film motor is a synchronous motor with six phases. The slider(top film) is excited by two-phase ac voltage and the stator (bottom film) is excited by four-phase ac voltage.

sets of three-phase ac voltage in previous studies [13], [14]. The electrode pitches of the stator $p_{\rm st}$ and the slider $p_{\rm sl}$ are 200 μm and 400 μm, respectively. Both of their electrode areas represent 46 mm $\times$ 30 mm. In order to decrease the influence of friction between the slider and the stator, we scatter glass beads of 7 μm in diameter between the two films. The more detailed explanation and analysis in relation to the electrostatic film motor will be described in Sections III-A, III-B, and III-C. It is important to note that the electrostatic film motor usually works under a high voltage so as to obtain a practical electrostatic force. In general, the electrostatic field between the films will exceed the maximum field strength of air under such high voltage so that a resultant discharge occurs in the air gap. Thus, dielectric liquid (e.g., Fluorinert FC-77, 3M Comapny) should be injected into the gap between the stator and slider films in order to avoid the undesirable discharge.

A. Principle

The electrostatic motor is a synchronous motor with six phases. It is driven by a two-phase ac voltage and a four-phase ac voltage, which are, respectively, imposed upon two-phase electrodes in the slider and four-phase electrodes in the stator (see Fig. 3). The driving voltages can be expressed in a vector from as

$$\begin{aligned}\mathbf{v} &= [v_1, v_2, v_3, v_4, v_5, v_6]^T \\ &= \Bigg[v_{\rm st}\ \sin(\omega_{\rm st}t), v_{\rm st}\ \sin\left(\omega_{\rm st}t+\frac{\pi}{2}\right), \\ &\quad v_{\rm st}\ \sin(\omega_{\rm st}t+\pi), v_{\rm st}\ \sin\left(\omega_{\rm st}t-\frac{\pi}{2}\right), \\ &\quad v_{\rm sl}\ \sin(\omega_{\rm sl}t), v_{\rm sl}\ \sin(\omega_{\rm sl}t+\pi)\Bigg]^T \\ &= [v_{\rm st}\ \sin(\phi_{\rm st}), v_{\rm st}\ \cos(\phi_{\rm st}), -v_{\rm st}\ \sin(\phi_{\rm st}), \\ &\quad -v_{\rm st}\ \cos(\phi_{\rm st}), v_{\rm sl}\ \sin(\phi_{\rm sl}), -v_{\rm sl}\ \sin(\phi_{\rm sl})]^T \end{aligned} \quad (1)$$

where v_1–v_6 represent the voltages applied to the corresponding six electrodes. $v_{\rm st}$, $v_{\rm sl}$, $\omega_{\rm st}$, $\omega_{\rm sl}$, $\phi_{\rm st}$, and $\phi_{\rm sl}$, respectively, represent voltage amplitudes, angular frequencies, and phases of the basic ac voltages applied to the stator and slider. Fig. 4. illustrates the electric potential distribution that are excited by the two- to four-phase ac voltages on the stator and slider.

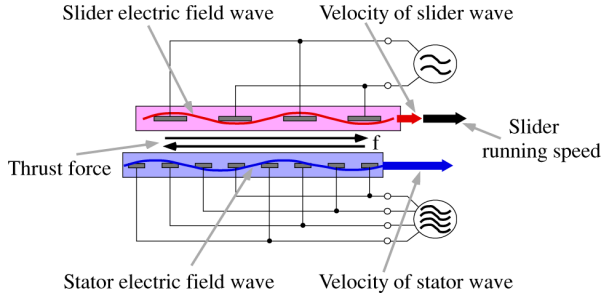


Fig. 4. Operation principle of an electrostatic film motor: the interaction between stator electric field wave and slider electric field wave generates the electrostatic force to drive the slider.

In our study, we have theoretically analyzed the thrust force of the two- to four-phase electrostatic film motor by using a six-terminal capacitance network model [14] [27]. As a result, the thrust force f in the driving direction of the electrostatic film motor can be trivially computed as

$$\begin{aligned} f &= Kv_{\rm st}v_{\rm sl}\ \sin(\phi_{\rm sl})\ \cos(\phi_{\rm st}-\theta) \\ &= Kv_{\rm st}v_{\rm sl}[\sin(\phi_{\rm sl}+\phi_{\rm st}-\theta) \\ &\quad +\sin(\phi_{\rm sl}-\phi_{\rm st}+\theta)]. \end{aligned} \quad (2)$$

Here, K is a coefficient that has relations with the capacitances among each electrodes, θ represents the electrical angle expression of the relative position of slider electrodes and is equal to $2\pi/2p_{\rm sl}x$ where x is the slider position. It should be mentioned that in the two- to four-phase electrostatic film motor, the ac driving voltage is required to be high frequency (e.g., 500 Hz or more). In the case of the high-frequency voltage, the component of $\omega_{\rm sl}+\omega_{\rm st}-\omega$ is too large to influence motion of the slider due to the fact that the actuator can hardly mechanically respond to the high-frequency signal. Ignoring the minute vibration, (2) can be simplified to be the following equation while the high-frequency ac signals are applied

$$f = Kv_{\rm st}v_{\rm sl}\ \sin(\phi_{\rm sl}-\phi_{\rm st}+\theta). \quad (3)$$

By supposing equilibrium, we obtain the relation between the thrust force f and the external force f_0, given by

$$f_0 = f = Kv_{\rm st}v_{\rm sl}\ \sin(\phi_{\rm sl}-\phi_{\rm st}+\theta). \quad (4)$$

Here, f_0 must satisfy the condition

$$-Kv_{\rm st}v_{\rm sl} \le f_0 \le Kv_{\rm st}v_{\rm sl} \quad (5)$$

to prevent the actuator from stepping out. On second thoughts, (4) can be rewritten as

$$\phi_{\rm sl}-\phi_{\rm st}+\theta = \sin^{-1}\frac{f_0}{Kv_{\rm st}v_{\rm sl}}. \quad (6)$$

If we consider external force f_0 and voltage amplitudes $v_{\rm st}, v_{\rm sl}$ to be constants, then differentiating (6) with respect to time t yields

$$\omega_{\rm sl}-\omega_{\rm st}+\omega = 0. \quad (7)$$

Since the relation between the electrode pitches of the stator and slider is $p_{\rm sl}=2p_{\rm st}$, and moreover,

$$\omega = \frac{\pi}{p_{\rm sl}}u, \qquad \omega_{\rm sl} = \frac{\pi}{p_{\rm sl}}u_{\rm sl}, \qquad \omega_{\rm st} = \frac{\pi}{2p_{\rm st}}u_{\rm st} \quad (8)$$

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we obtain the following relation by substituting them into (7)

$$u = u_{\rm st} - u_{\rm sl} \tag{9}$$

where u is the slider running speed, and $u_{\rm st}, u_{\rm sl}$, respectively, denote the velocities of electric field waves on the stator and slider. Expressing (9) with angular frequencies $\omega_{\rm st}$ and $\omega_{\rm sl}$, we obtain:

$$u = u_{\rm st} - u_{\rm sl} = \frac{2p_{\rm st}\omega_{\rm st} - p_{\rm sl}\omega_{\rm sl}}{\pi}. \tag{10}$$

Once the difference of angular frequencies applied to the stator and slider is modulated, the slider running speed is varied. Thus, the moving distance of the slider S during the operating time T is defined as

$$S = uT = \frac{2p_{\rm st}\omega_{\rm st} - p_{\rm sl}\omega_{\rm sl}}{\pi}T. \tag{11}$$

If the angular frequencies are expressed as a function of time t,[1] (11) can be rewritten as an integral format

$$S = \int_0^T \left[\frac{2p_{\rm st}}{\pi}\omega_{\rm st}(t) - \frac{p_{\rm sl}}{\pi}\omega_{\rm sl}(t)\right] dt. \tag{12}$$

As outlined in (10) and (11), the moving distance of the slider is proportional to the difference of $\omega_{\rm st}$ and $\omega_{\rm sl}$ during the operating time T if $\omega_{\rm st}$ and $\omega_{\rm sl}$ are constant. When $\omega_{\rm st} > \omega_{\rm sl}$, the slider is moving along a certain direction in order to make up for the velocity difference between the electric field waves on the stator and slider. Conversely, the slider moves toward the opposite direction when $\omega_{\rm st} < \omega_{\rm sl}$.

B. Open-Loop Control Method

As formulated in (10), it is possible to adjust the slider running speed by modulating the angular frequencies of slider and stator. More concretely, if the difference between the angular frequencies of slider and stator is constant, the slider will move with a uniform speed. For the case in which the difference between both gradually increases, the slider accelerates. In contrast, the slider will slow down when the difference between the angular frequencies of slider and stator decreases. Of course, the slider stops moving when the difference becomes zero.

To imitate muscle-like activation pattern, we should accomplish the reciprocating motion on the electrostatic film motor. Moreover, a smooth speed modulation is required because the electrostatic linear motor will step out if the slider running speed is changed excessively. Therefore, we engage the following open-loop control method to realize the smooth reciprocating motion

$$\begin{aligned} \omega_{\rm st}(t) &= \omega_{\rm const} \\ \omega_{\rm sl}(t) &= A\sin\left(\frac{2\pi}{T_{\rm sw}}t\right) + \omega_{\rm const} \end{aligned} \tag{13}$$

where $\omega_{\rm const}$ is constant angular frequency, A is the amplitude of angular frequency sweeping, and $T_{\rm sw}$ represents the operating time (i.e., sweeping period). Substituting frequencies for angular frequencies, (13) can be rewritten as follows:

$$\begin{aligned} f_{\rm st}(t) &= f_{\rm const} \\ f_{\rm sl}(t) &= A^*\sin\left(\frac{2\pi}{T_{\rm sw}}t\right) + f_{\rm const}. \end{aligned} \tag{14}$$

Here, A^* is the amplitude of frequency sweeping, i.e., $A^* = A/2\pi$. As a result, the six-phase driving ac voltages applied to the electrostatic film motor are given by

$$\begin{aligned} \mathbf{v} &= [v_1, v_2, v_3, v_4, v_5, v_6]^T \\ &= \Big[v_{\rm st}\ \sin(2\pi f_{\rm st}(t)t), v_{\rm st}\ \cos(2\pi f_{\rm st}(t)t), \\ &\quad -v_{\rm st}\ \sin(2\pi f_{\rm st}(t)t), -v_{\rm st}\ \cos(2\pi f_{\rm st}(t)t), \\ &\quad v_{\rm sl}\ \sin(2\pi f_{\rm sl}(t)t), -v_{\rm sl}\ \sin(2\pi f_{\rm sl}(t)t)\Big]^T. \end{aligned} \tag{15}$$

This control scheme corresponds to a feedforward position control since the open-loop controller does not engage any sensory feedback when it adjusts the moving distance of the slider. In the next section, we confirm the feasibility and validity of the open-loop controller in the actual application via prototype tests.

C. Prototype Testing

We first fixed the stator film on a base. Next, we placed the slider film on the stator film. Dielectric liquid was injected into the gap between the stator and slider films to avoid the undesirable discharge. The required two-phase and four-phase sinusoid signals were generated by two commercial function generators (WF1946A, NF Company). Through commercial high-voltage amplifiers (model 609C, Trek Company) with a gain of 1000, the two-phase and four-phase high-voltage signals capable of driving the electrostatic film motor were, respectively, applied to the slider and the stator. Finally, we accomplished smooth reciprocating motion according to the control method discussed in Section III-B.

The reciprocating motion of the electrostatic film motor was observed by using an electrooptical displacement transducer (Model 200, Zimmer Company). The electrical signals from the optical displacement transducer were sent to an oscilloscope so as to measure and record the data.

Fig. 5 shows the theoretical and the measured slider moving distance for a single sweeping. The frequency of the stator [i.e., $f_{\rm st}$ in (14) was fixed at 500 Hz. The sweeping span of the frequency of the slider was $\pm$25 Hz around the fixed frequency of the stator. The sweeping period [i.e., $T_{\rm sw}$ in (14)] was set as 0.6 s. The driving voltages of 1 $\rm kV_{0-p}$ were, respectively, applied to the stator and the slider. The experimental data show that the moving distance of the slider matches the model predictions closely. We note that the proposed open-loop modulating method has good response and is adequate to the general control without high precision. In this case, the maximum thrust force and power consumption of the electrostatic film motor were approximately 0.3 N and 0.6 W, respectively.

[1]For instance, the frequency is continuously modulated during a constant period when we use the electrostatic motor to provide propulsion for the designed fish-like robot.

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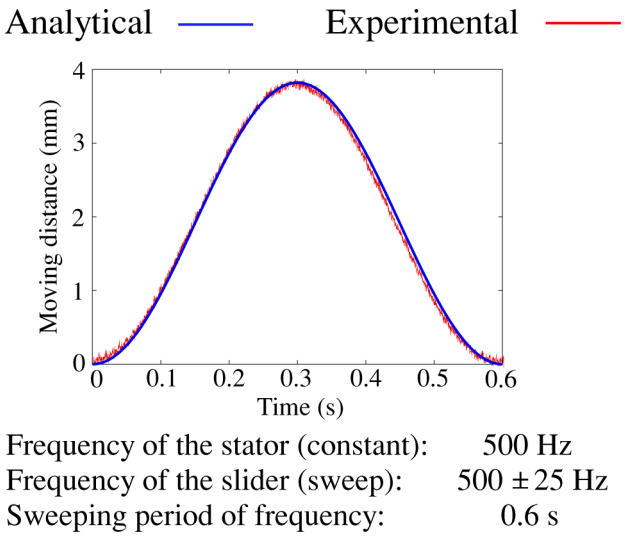


Fig. 5. Theoretical and experimental slider moving distance during one sweeping period (i.e., 0.6 s) when the electrostatic film motor is reciprocating.

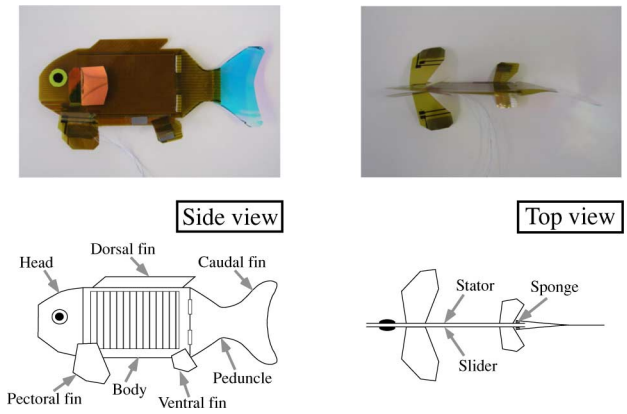


Fig. 6. Photograph (top) and schematic structure (bottom) of the Seidengyo I robotic fish. The size of the designed robot is 130 mm × 50 mm × 55 mm and its total weight is only 5 g.

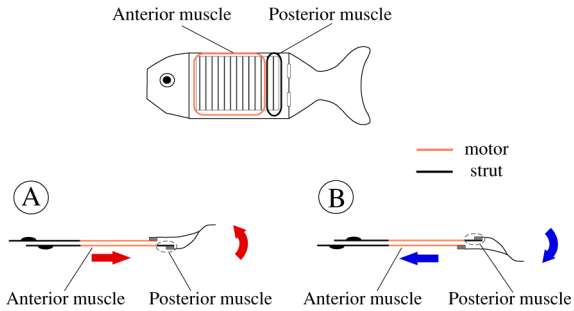


Fig. 7. Electrostatic film motor and a power transmission system for Seidengyo I. A fish swimming pattern in muscle function is applied to the system design. As the oscillation of the caudal fin continues,the posterior film of the motor changes its role between a motor and a strut.

IV. Design Features and Control of the Seidengyo I Robotic Fish

The first step in designing a robotic fish is to select a desired caudal fin motion. In our design, a lunate caudal fin undergoing small amplitude oscillation is chosen to mimic biological propulsion. Full and Koditschek [15] indicated that locomotion results from complex, high-dimensional, nonlinear, dynamically coupled interactions between an organism and its environment. Videler [18] stated further that the actual swimming locomotion is caused by the interaction between the body and the surrounding water. Hu and his coworkers [23] found that their fish-like robot with a passive and flexible caudal fin is capable of generating the fish-like undulating motion to propel itself forward. Motivated by aforementioned points of view, we conclude that when fishes cruise at high speeds, passive dynamic self-stabilization from a feedforward and tuned mechanical system can reject rapid perturbations and simplify control. In fact, there is no carefully controlled fin oscillation or noticeable changes in undulating pattern. Therefore, even a simple feedforward or open-loop control is capable of stabilizing the caudal-fin-based swimming if only the robotic fish owns enough self-stabilizing properties to immediately respond to perturbations without sensory feedback.

A. Mechanical Design

Our first-generation prototype robotic fish, referred to as Seidengyo[2] I, was fabricated for the simple task of stable swimming. Thus, it was designed with a lunate flexible caudal fin and a relatively narrow peduncle. The schematic structure and photograph of the Seidengyo I robotic fish are shown in Fig. 6. The body of the robot consists of the stator and slider films. Namely, the entire fish body can be considered to be an electrostatic film motor. Although the dorsal, pectoral, and ventral fins are equipped, they are inactive and used to maintain balance in liquid. Seidengyo I is 130 mm in length, 50 mm in width, and 55 mm in height. The total weight is only 5 g. To the best of our knowledge, Seidengyo I is simplest and lightest in the fish-like robot literature.

A muscle function similar to living fish (see Fig. 1) has been designed in Seidengyo I, as shown in Fig. 7. The overlapped parts of the stator and slider are regarded as anterior muscles, and the projecting part of the stator or slider is regarded as posterior muscles. In the case of A, the slider film moves backward and the projecting part of the slider film appears. Simultaneously, the actuator energy is transmitted to the caudal fin via the projecting part, and a part of kinetic energy is stored or absorbed by the prominence due to its viscoelasticity. So, the projecting part acts as a strut, as illustrated in Fig. 1. In the case of B, the projecting part of the stator film appears as the slider film moves forward. Conversely, the former projecting part of the slider is transformed into the overlapped part and acts as a motor. This configuration that permits reciprocating power to be converted to periodic oscillation of a flexible caudal fin is based on the studies of fish swimming patterns in muscle function. In addition, one major concern is power density, for which it is desirable to use a light, thin, and flexible electrostatic film motor

[2]It is named after Japanese words meaning electrostatic (seiden) and fish (gyo).

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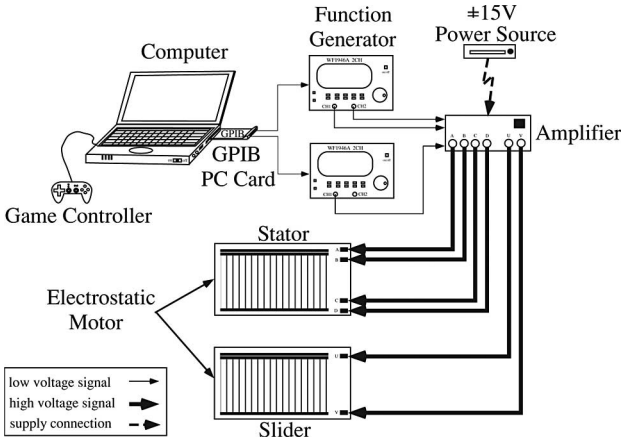


Fig. 8. Logical connection among each control component. Through a high-voltage amplifier, a three-phase driving ac signal is changed as a six-phase driving ac signal.

as the primary actuation source. For large robots, the actuator energy can be stored elastically and periodically released by using a sophisticated spring-damper mechanism. At the scale of Seidengyo I, however, it becomes easier to store kinetic energy since it takes advantage of the elastic potential of the flexible film alone without redundant mechanisms.

B. Swimming Control

Control problems for a robotic fish include various topics (e.g., trajectory planning, tracking trajectory, efficient swimming, high maneuverability, etc.). This study is a challenge to an application of a unique actuator (i.e., electrostatic film motor) in robotic fish. As the first stage of the study, we will focus on problems relating to generation and adjustment of swimming. In this section, drawing inspiration from fish, we propose several open-loop control strategies to realize the fish-like motion.

1) Controller Description: As described in aforementioned sections, the electrostatic motor essentially needs two- to four-phase driving ac signals (i.e., six driving ac signals) for the operation. Fortunately, it is easy to increase a new driving ac signal as long as we reverse a certain signal in general. Thus, three driving ac signals can be increased to two- to four-phase driving ac signals by turning around to the opposite direction. As shown in Fig. 8., we utilize two function generators (WF1946A, NF Company) to provide the three driving ac signals. The one provides two signals with 90° phase difference so as to generate the four-phase driving ac signals. The other provides one signal so as to generate the two-phase driving ac signals. Through a high-voltage amplifier that is composed of three sets of power operational amplifiers (OPA548, Burr-Brown Company) and transformer circuits, the four-phase and two-phase high ac voltages are fed to the stator and the slider, respectively. The central component of the control circuit is a laptop computer with General Purpose Interface Bus (GPIB) (IEEE 488) interface card. The computer sends out signal commands to function generators through the GPIB interface.

All control programs are developed with a visual programming language, LabVIEW (National Instruments Company). The control program sends the control commands concerning the frequency, initial phase, and amplitude of each driving ac signal through the GPIB interface. The function generators mainly produce two V_{p-p} sinusoidal wave signals according to the control command from the computer. Since the amplifier has a gain of 1000, the ac voltages applied to the electrostatic motor can be controlled at 2 kV_{p-p}.

2) Generation of Swimming Motion With Flexible Caudal Fin: In Seidengyo I, the open-loop control method described in Section III-B is used to generate the required propulsion of fish-like swimming. More concretely, the electrostatic film motor first generates reciprocating motion in front and rear by modulating the frequency of sine waves applied to the slider. Then, the reciprocating power is converted to periodic oscillations of a flexible caudal fin by the power transmission system discussed in Section IV-A.

In the swimming control, the constant frequency $f_{\rm const}$ always maintains 500 Hz. The amplitude of frequency sweeping A^* is set as 25 Hz. The period of frequency sweeping $T_{\rm sw}$ is approximately 0.6 s. As a result, the slider runs within $\pm$20 mm/s speed and produces oscillating motion of the caudal fin to propel the robot forward.

Fig. 9 illustrates the oscillation of the caudal peduncle and fin during a single frequency modulation period. It should be mentioned here that the oscillating motion only generates a C-shape bend of the caudal fin in air, but in water, it can generate correct fish-like undulating motion. Since the flexible caudal fin is pushed through the water, its top is inversed from the driven C-shape bend by the water resistance. This shape inversion undulates the caudal fin and provides sufficient propulsion. The fact could provide a rational and possible explanation of why our fish-like robot can achieve good swimming performance only using an active joint and a very simple open-loop control method. It is in agreement with the recent research from biomechanics, which shows that fish make use of passive mechanical properties, most notably compliance, to enhance the effectiveness of their muscles [17].

3) Turning Motion Strategy: In order to achieve better fish-like maneuverability, we design a strategy that takes advantage of the characteristic of electrostatic film motor to control the left/right turning. It is well known that many fish continuously change the body shapes, especially bend the caudal fins, to accomplish the turning. In the same way, such continuously changing or bending can be achieved in Seidengyo I by adjusting the neutral point of the caudal fin. A turning motion strategy that takes these inspirations into consideration using initial phase and frequency modulations of driving ac voltages is as follows:

$$\begin{aligned}\mathbf{v} &= [v_1, v_2, v_3, v_4, v_5, v_6]^T \\ &= [v_{\rm st}\ \sin(2\pi f_{\rm st}(t)t), v_{\rm st}\ \cos(2\pi f_{\rm st}(t)t), \\ &\quad -v_{\rm st}\ \sin(2\pi f_{\rm st}(t)t), -v_{\rm st}\ \cos(2\pi f_{\rm st}(t)t), \\ &\quad v_{\rm sl}\ \sin(2\pi f_{\rm sl}(t)t+\psi), -v_{\rm sl}\ \sin(2\pi f_{\rm sl}(t)t+\psi)]^T \end{aligned} \quad (16)$$

where $f_{\rm st}$ and $f_{\rm sl}$ are still defined as (14). ψ represents initial phase and is considered to be a shift of the neutral point. The

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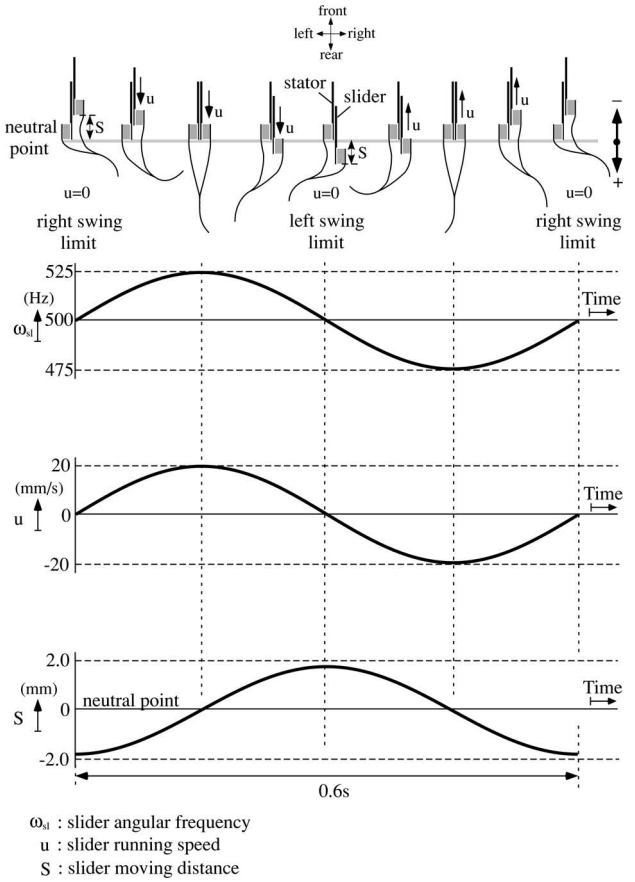


Fig. 9. Time history of the oscillation of the caudal peduncle and fin. During one frequency modulation period (i.e., 0.6 s), the caudal fin generates correct fish-like undulating motion under the operation of the electrostatic film motor.

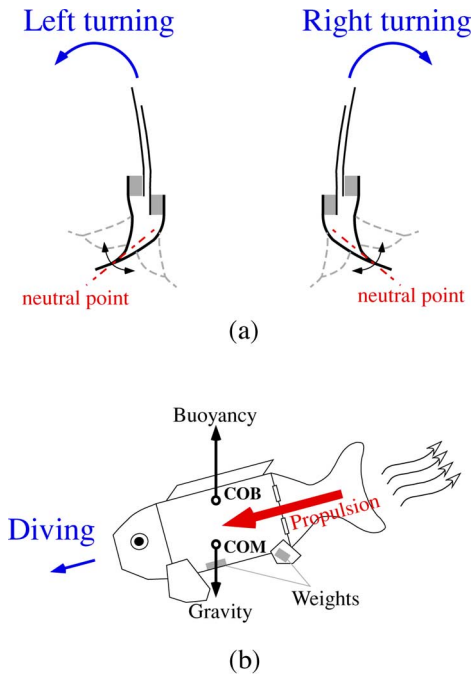


Fig. 10. Schematic diagrams of turning and diving motion strategies. (a) Robotic fish shifts the neutral point of the caudal fin oscillation to the left or right side so as to achieve the corresponding turning motion. (b) Robotic fishuses the downward propulsion to realize gradual diving motion.

shifting algorithm of the neutral point is summarized as follows:

$$\psi = \begin{cases} \pm\lambda t, & 0 < t \le T_{\rm sh} \\ \pm\lambda T_{\rm sh}, & t > T_{\rm sh} \end{cases} \quad (R:-;L:+) \qquad (17)$$

where λ denotes the gain constant that is fixed at 2000°/s in Seidengyo I, and $T_{\rm sh}$ is the shifting period.[3] To prevent the actuator from stepping out, we set the shifting period as 1.0 s. R and L represent right and left turning.

Intuitively, this simple turning motion strategy can be described as two main steps: offset and oscillate [see Fig. 10(a)]. First, the initial phases of sinusoidal waves applied to the slider are modulated during 1.0 s so as to produce the offset between the stator and slider. As a result, the neutral point of the caudal fin is adjusted toward left or right side as the initial phase ψ is set at the positive or negative value. Next, the robot operates the frequency modulation mentioned in Section IV-B2 to realize the oscillation of the caudal fin around the adjusted neutral point. In the practical operation, the two steps are almost simultaneously implemented. As a result, Seidengyo I accomplishes the left/right turning like a real fish.

[3]The changing time when the neutral point of the caudal fin is shifted from the central location to the right or left location.

4) Diving Motion Strategy: In fact, many fish are essentially open-loop unstable. For instance, most dead fish tend to float belly-up, suggesting that the center of buoyancy (COB) lies below the center of mass (COM). Thus, most fish can be considered to be inverted pendulums and must use feedback control (e.g., adjustment of air bladder and motion of fins) to remain upright. In addition, pitch instabilities of fish are also common because the COB is typically located fore and hind of the COM [28]. Since this study is an attempt on applying a unique electrostatic film motor and a simple swimming control method to a robotic fish, the sophisticated attitude and position control are beyond the scope of this paper.

In this study, we utilize the adjustment of centroid to keep up upright posture and pitch stabilities. As shown in Fig. 10(b), we attach the suitable weights to the abdomen and ventral fins so that the COM lies below the COB and the gravity is somewhat less than the buoyancy. This configuration makes Seidengyo I slowly rise to the surface with the upright posture in lateral plane when the caudal fin stops oscillating (i.e., no propulsion is provided).

Furthermore, we incline Seidengyo I's head down by using the adjustment of centroid [see Fig. 10(b)]. The slantwise fundamental posture in sagittal plane facilitates the realization of diving motion. In Seidengyo I, only the caudal fin can provide propulsion due to a single actuator. Using the downward propulsion makes Seidengyo I gradually dive down as it ceaselessly cruises in water. By providing the proper downward propulsion or not, the Seidengyo I robotic fish is capable of freely cruising at the optional depth of water.

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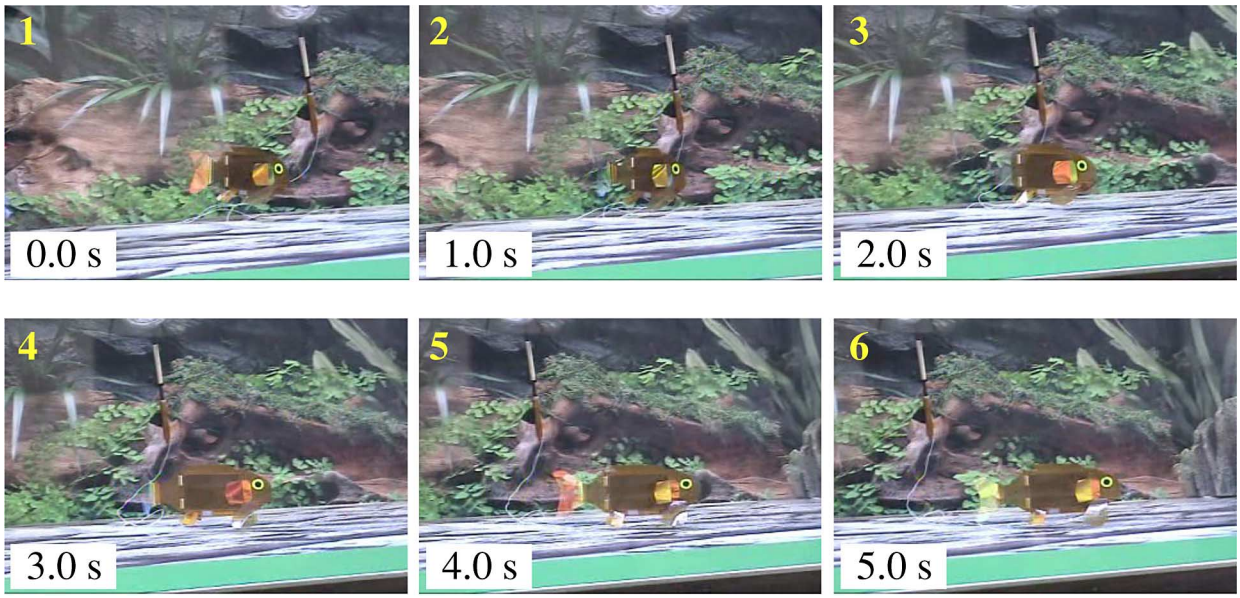


Fig. 11. Snapshots of the Seidengyo I robotic fish swimming in dielectric liquid, i.e., Fluorinert FC-77 (3M, Company).

V. Experiments of Seidengyo I and Discussion

A. Equipment and Method of Robot Measurement

To validate the original designs, we used Seidengyo I to achieve its swimming in an aquarium (see Fig. 11). The length, width, and depth of the aquarium were, respectively, 750, 350, and 450 mm. Since water is conductive, the electrostatic film motor cannot work if the robotic fish directly swims in water. Thus, we infused dielectric liquid, Fluorinert FC-77 (3M Company, density: 1780 kg/m^2, kinematic viscosity: 0.8 cSt), into the aquarium. A network/spectrum/ impedance analyzer (Agilent 4395, Agilent Technologies Company) measured the phase variations in the driving ac signals between the stator and slider at 500 Hz measuring frequency.

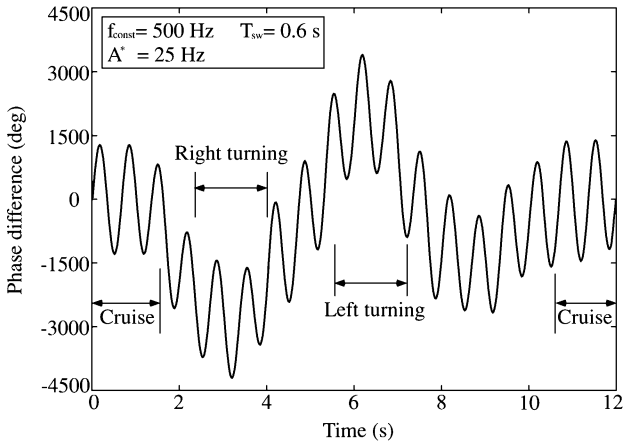


Fig. 12. Phase variations of the driving ac signals between the stator and slider for cruising and turning motions of Seidengyo I.

B. Robot Measurement Results

Fig. 12 shows the phase variations of the driving ac signals between the stator and slider in a swimming program. In the case of the swimming program, Seidengyo I cruised straight for 1.8 s, then made a right turn until approximately 4.2 s. As the robotic fish turned its body rightward, the phase difference was shifted to the negative value, as described in Section IV-B3. In contrast, the phase difference was shifted to the larger positive value while Seidengyo I turned leftward. After approximately 10.4 s, Seidengyo I cruised straight afresh. The cruise speed of Seidengyo I was approximately 0.018 m/s and its turning speed of 0.008 m/s was about 55% slower than the cruise speed.

Fig. 13 shows the phase variations of the driving ac signals between the stator and slider for swimming speed transition of Seidengyo I. The constant frequency $f_{\rm const}$ was also fixed at 500 Hz. For the duration (1) as plotted in Fig. 13, the sweep period T_{sw1} was 1.2 s and the amplitude of frequency sweeping A_1^* was fix at 5 Hz. At that time, Seidengyo I swam at a slower speed of about 0.012 m/s. For the duration (2), the sweep period T_{sw2} and the amplitude of frequency sweeping A_2^* were, respectively, modulated to be 0.6 s and 10 Hz. Then, Seidengyo I's swimming speed decreased to approximately 0.008 m/s. It is important to note that although the phase transition was hardly remarkably smooth, the electrostatic film motor still enabled the caudal fin to oscillate periodically.

In our experiments, it is detected that Seidengyo I's swimming speed increases as the oscillating amplitude of the caudal fin is increased. In fact, the oscillating amplitude of the caudal fin is in direct proportion to the moving distance of the slider of the electrostatic film motor. As formulated in (12), the moving distance of the slider is decided by the phase difference between driving ac signals applied to the stator and slider. Furthermore, it is also detected that the swimming speed is in inverse proportion to the oscillating frequency in the case of the same oscillating amplitude.

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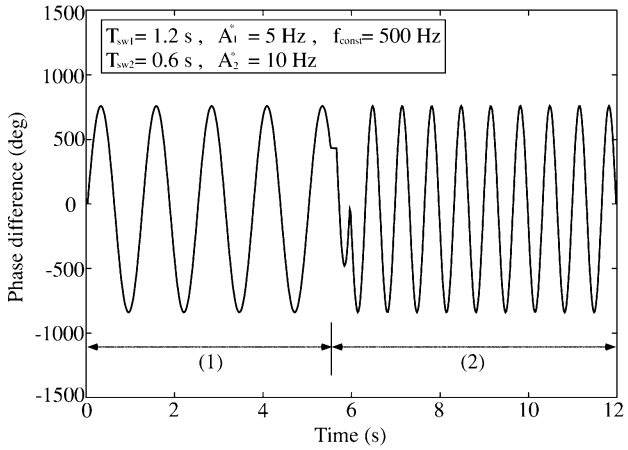


Fig. 13. Phase variations of the driving ac signals between the stator and slider for swimming speed transition of Seidengyo I.

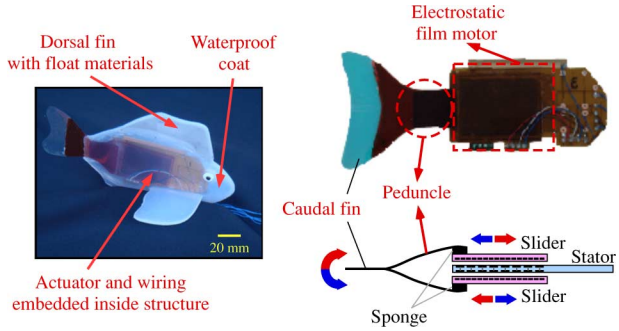


Fig. 14. Seidengyo II, a dynamically stable robotic fish actuated by a three-layer-structure electrostatic film motor. The original prototype is capable of swimming in water because of the shaped waterproof coat.

C. Limitations of Seidengyo I

Although the designed mechanical and control systems of Seidengyo I was capable of achieving fish-like swimming, the robotic fish could only cruise in dielectric liquid due to the fact that the electrostatic film motor cannot operate normally in water because water is a conductor of electricity that destroys the electrostatic field between the stator and the slider. In addition, the bodies of the thunniform swimmers are well streamline to significantly reduce pressure drag [16]. In spite of the power of the caudal thrusts, the streamline body and rational mass distribution facilitate effectively minimizing the recoil forces and keeping from sideslip. Therefore, it becomes indispensable to our robotic fish that the actuator is embedded in a waterproof coat that is fabricated to be enough flexible and well streamline fish-shaped body.

On the other hand, the controller of Seidengyo I mainly depended on a laptop computer with the GPIB interface card to generate various control commands. It is adequate to cope with the feedforward mode transitions (e.g., changing cruise mode to turning mode, beginning cruise mode from stop mode, and so on), as implemented in this study. In the case of the mode transitions, each motion mode is switched by using the corresponding button on an game controller. But, when we used the swing of the analogue joystick on the game controller to synchronously manipulate the oscillation of the caudal fin, the constructed control system could not smoothly drive the actuator to generate the required oscillating motion. Here, the relation between the swing angle of the analogue joystick α and the moving distance of the slider S is defined as

$$\alpha \propto S. \tag{18}$$

More concretely, we apply the six-phase driving ac voltages to the stator and slider films, as formulated in (16). The frequencies $f_{\rm st}$,$f_{\rm sl}$, and the initial phase ψ are defined as

$$\begin{cases} f_{\rm st}(t) = f_{\rm const} \\ f_{\rm sl}(t) = f_{\rm const} \\ \psi = \eta\alpha \end{cases} \tag{19}$$

where η is the synchronous operation coefficient of 40. The swing angle α denotes positive values when the joystick is swung over to left side and vice versa, when swung over to the right side. It is desired that along with the swing of the joystick, the slider synchronously moves so as to generate synchronous oscillation of the caudal fin. It is obvious that the control system of Seidengyo I is inadequate to generate proper driving ac signals in time along with the swing of the joystick. This indicates that the control system of Seidengyo I lacks the real-time ability and maneuverability needed to operate in autonomous swimming.

VI. Development of Improved Seidengyo II

As discussed previously, the limiting factors in the design of Seidengyo I have been its incompatibility with water, lower maneuverability, nonreal-time control, and large-sized controller scheme. Thus, our next work focuses on solutions of these problems. In this section, we present an improved version of the electrostatic fish, referred to as Seidengyo II. The incorporation of a waterproof coat, a redesigned electrostatic film motor (i.e., three-layer-structure actuator), and a compact and versatile controller have enabled Seidengyo II to swim in water at speeds of over 0.04 m/s. Despite changes in the body shape and force generation mechanism, we show that by effectively making use of the electrostatic film motor, a fish-like muscle function is accomplished and the fast, self-stabilizing swimming behavior of the robot is preserved.

A. Mechanical Design of Seidengyo II

To tone up the power of the artificial muscle, we redesign the power transmission system of Seidengyo II, as illustrated in Fig. 14. The three-layer-structure actuator consists of one stator film and two slider films where the stator is fixed within the waterproof coat and two sliders are fitted on alternate sides of the stator. As shown in Fig. 14, the two sliders are attached to the peduncle films. By respectively driving the two sliders to generate two completely inverse but synchronous reciprocating motions, the force generation mechanism similar to that described in Section IV-A has been achieved. It should be mentioned that the former stator and slider films have only a single-sided working area due to the limiting manufacture technique relating to FPCs. In this system, we interchange the stator and the slider

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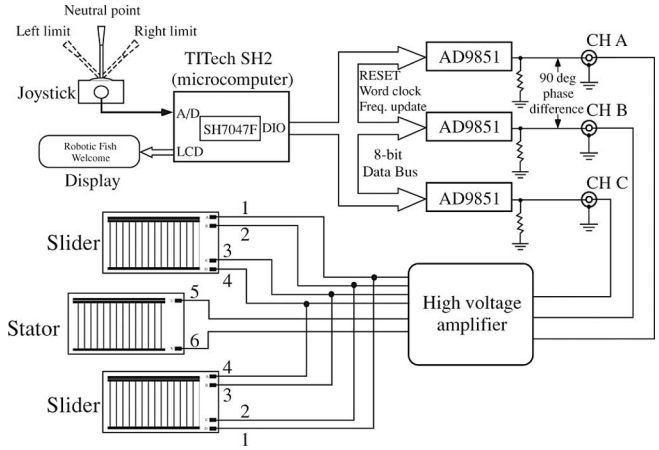


Fig. 15. Schematic diagram of the control system of Seidengyo II. The whole control tasks are implemented every 1 ms in order to realize real-time control.

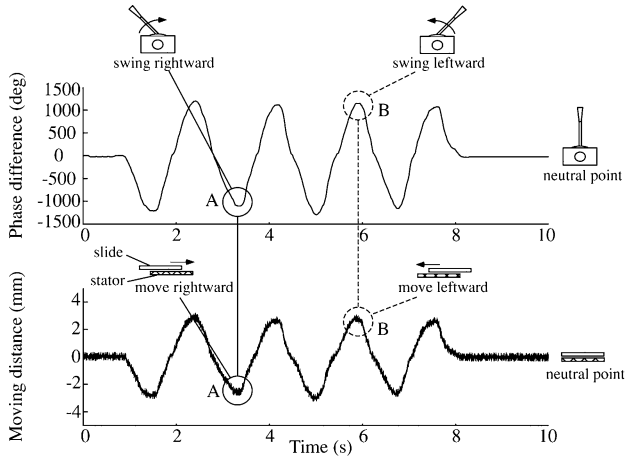


Fig. 16. Test results of the constructed control system. The phase difference between the control signals of the stator and slider is modulated by using a joystick. Simultaneously, the electrostatic film motor is synchronously driven as the phase difference changes.

in the actuator where the stator film has two-phase electrodes and the slider film has four-phase electrodes. This configuration is different from that of Seidengyo I and convenient for obtaining a stator film with double-sided working area by a dual-side manufacture technique. In general, it is impractical to produce a four-phase electrode FPC film with a double-sided working area.

Furthermore, we shape a waterproof coat out of the silicon material so that Seidengyo II can directly cruise in water. As shown in Fig. 14, the waterproof coat is designed to be a fish-like shape. The actuator and wiring are embedded within the fish-shape waterproof coat. To avoid the undesirable discharge, we inject silicon oil into the waterproof coat. Fortunately, since the waterproof coat is supported by the silicon oil, the body of Seidengyo II forms a streamline shape and its stiffness is approximately 3.2 N/m. In addition, the pectoral fin is designed to be like a wing to provide balance, and the dorsal fin is made from float materials. The buoyancy of Seidengyo II in water can be modulated by expanding float materials in advance.

The body size of Seidengyo II is 175 × 95 × 100 mm and its total weight is approximately 85 g. Even though Seidengyo II is larger and heavier than Seidengyo I, its stability and maneuverability still do not lag behind those of Seidengyo I and even exceed them. This suggests again that appropriate biomimetic designs facilitate improving the locomotion performance of robots.

B. Improvement of Control Systems

In the controller of Seidengyo II, we employed a programmable waveform generator (AD9851, Analog Devices Company) to synthesize the high-frequency sinusoidal signals and a specially made microcomputer referred to as TITechSH2 (Hibot Company) that includes a 50 MHz SH7047F CPU (Renesas Company), 16-channel 10-bit A/D converters, 22-bit digital I/O ports, a liquid-crystal display (LCD) display output, two-port encoders, and a serial port.

Fig. 15 schematically shows the control system of Seidengyo II. First, the TITechSH2 commands three AD9851 synthesizers to generate sinusoidal signals with frequencies of 500 Hz (i.e., CH A, CH B, and CH C in Fig. 15). It is noted that a 90° phase difference is fixed between the CH A and CH B signals. Next, these ac signals are amplified to be six ac high voltages through a high-voltage amplifier made by ourselves. The driving ac high voltages of

$$\begin{aligned}\mathbf{v} &= [v_1, v_2, v_3, v_4, v_5, v_6]^T \\ &= \Big[\, v_{\rm sl}\,\sin(2\pi f_{\rm sl}(t)t), v_{\rm sl}\,\cos(2\pi f_{\rm sl}(t)t), \\ &\quad -v_{\rm sl}\,\sin(2\pi f_{\rm sl}(t)t), -v_{\rm sl}\,\cos(2\pi f_{\rm sl}(t)t), \\ &\quad v_{\rm st}\,\sin(2\pi f_{\rm st}(t)t+\psi), -v_{\rm st}\,\sin(2\pi f_{\rm st}(t)t+\psi)\,\Big]^T \end{aligned} \quad (20)$$

are then applied to the electrostatic film motor as actuator power and control signals. Here, we redefine $f_{\rm sl}$ and $f_{\rm st}$ as follows:

$$\begin{aligned} f_{\rm sl}(t) &= f_{\rm const} \\ f_{\rm st}(t) &= A^* \sin\left(\frac{2\pi}{T_{\rm sw}}t\right) + f_{\rm const}. \end{aligned} \quad (21)$$

The shifting algorithm of the neutral point identical with (17) is still utilized. In the three-layer-structure actuator, we only reverse the ac signal order of a slider in sequence so as to obtain the ac signals of the other slider, as shown in Fig. 15.

C. Experimental Study

To confirm the real-time ability and versatility of the constructed controller to easily compare with that of Seidengyo I, we again attempt to realize a synchronous operation approach as formulated in (18) and (19), in which the swing of the joystick is sampled at 1 ms interval by the TITechSH2 via the A/D converter. Based on the sampling data (i.e., swing angle value of the joystick), the TITechSH2 modulates the initial phase ψ of the sine wave signals applied to the stator. More concretely, the initial phase is set as 0° when the joystick is at neutral point. It is increased as soon as the joystick swings over to the right side.

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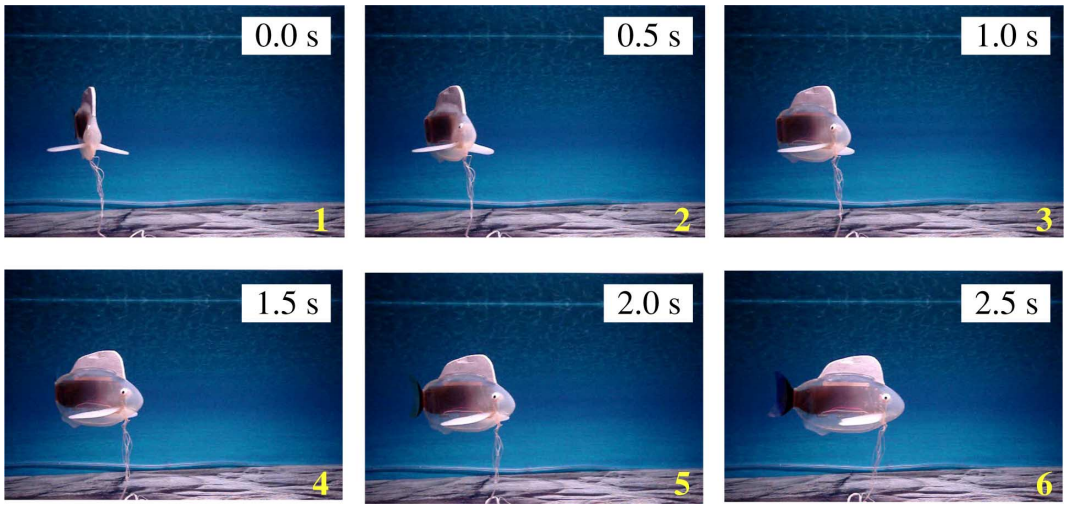


Fig. 17. Snapshots of the Seidengyo II robotic fish swimming in water. Seidengyo II implemented a fast C-shape sharp turning with an efficient turning motion strategy.

Conversely, when the joystick swings over to the left side, the initial phase is decreased. Since the maximum left/right limits of the swing angle of the joystick are +/− 45°, the corresponding left/right limits of the initial phase are fixed at +/− 1800°. Fig. 16 shows the test results of the synchronous operation controlling swimming of Seidengyo II. While a user swang the joystick in right and left direction, the initial phase of the control signal applied to the stator (i.e., CH C in Fig. 15) would be increased or decreased. Almost simultaneously, the electrostatic film motor was also driven according to the swing pattern of the joystick. As a result, the caudal fin of Seidengyo II generated oscillations to propel itself.

The swimming control strategies of Seidengyo II are almost identical with those of Seidengyo I except the turning motion strategy. Compared with Seidengyo I, Seidengyo II effectively minimized the recoil force in water so as to enhance the propulsion because of the streamline body. This advantage provides a possibility for using a more efficient turning motion strategy that is adopted by a great many thunniform swimmers. In the efficient turning motion strategy, the robotic fish only bends its caudal fin toward left or right side quickly so as to make the whole body be a C-shape. By utilizing the inertial force and interaction between the compliant body and the surrounding water, the robotic fish fulfills a C-shape turning. The turning motion strategy is considered only a caudal fin offset and no caudal fin oscillation, compared to that of Seidengyo I. Fig. 17 shows a set of snapshots where Seidengyo II implemented a fast and sharp turning for the duration of approximately 2.5 s.

VII. Conclusion and Future Work

We have built the first robotic fish with an electrostatic film motor, thus demonstrating that applying such a unique actuator to robotics is possible. The robotic fish relies on the dynamics of the flexible and high-power electrostatic film motor and passively compliant designs of body and caudal fin. We have proposed several simple swimming control strategies that rely on open-loop feedforward (i.e., no sensory feedback). The proposals resulted not only in swimming at average cruising speeds up to approximately 0.018 m/s but also in good fish-like mobility (i.e., sharp turning, diving, and countermarching). Our robotic fish provided an experimental instantiation of fundamental design and control principles for a new class of biomimetic robots with reduced mechanical complexity and power requirements.

Some deficits in Seidengyo I, including incompatibility with water, lack of maneuverability, and expensiveness of the controller, have been also addressed in this paper. The use of a waterproof coat, a versatile microcomputer, and a self-made high-voltage amplifier on Seidengyo II allowed us to solve aforementioned problems and enhance performance and adaptability. The difficulty in this study relating to two- to four-phase electrostatic film motor is that the high-voltage and high-frequency driving signals are required. More sophisticated detection and trial product of power generator is an area of ongoing work.

More generally, the prototypes presented in this paper begin to approximate the performance of an actual fish and are lighter, thinner, and more flexible than most previous robotic fish. In addition, the proposed design and control scheme concerning an electrostatic film motor is an example of an approach that is particularly well suited for small, flexible, biomimetic robots by requiring no expensive or sophisticated sensing or feedback.

Although the simple open-loop control scheme is sufficient for swimming in water, we still advocate challenging to large environmental perturbations, rapid maneuvering, and energy-efficient endurance swimming. Future work will continue on the understanding of hydrodynamics of the robotic fish and further introduce the bioinspired control strategy (e.g., neural feedback). As described previously, the control strategy based on biological concepts is capable of realizing more autonomous and stable swimming and obtaining more complex motion (e.g., preying and dodging). Finally, future work will also address to the development of a wireless robotic fish. For example, electromagnetic induction method can be utilized to get rid of power cables due to the fact that Seidengyo II's active space is assumed in a limited aquarium.

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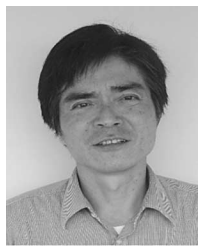
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